

TURKMENBASHI, Turkmenistan

WMO: 381412

Lat: 40.063N

Lon: 53.007E

Elev: 85

StdP: 100.31

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-5.0	-15.2	1.0	-1.8	-12.0	1.3	1.4	11.6	4.8	10.4	5.8	4.0	80	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	38.9	19.7	37.3	19.4	36.0	19.3	25.0	31.0	23.9	30.1	22.9	29.6	5.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.2	18.2	28.2	22.0	16.8	27.3	20.9	15.7	26.8	76.8	30.9	72.2	30.1	68.4	29.7	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.6	9.7	8.8	-10.9	41.2	4.2	1.4	-13.9	42.2	-16.4	43.0	-18.8	43.8	-21.8	44.9		
			-11.5	26.8	4.1	1.6	-14.5	28.0	-16.8	28.9	-19.1	29.8	-22.1	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	3.8	4.3	8.5	14.0	20.4	25.7	28.7	28.9	23.2	16.0	9.4	5.2
	DBStd	9.85	3.92	4.30	3.68	4.02	3.64	3.57	2.77	2.81	3.32	3.89	4.11	3.84
	HDD10.0	655	196	163	72	9	0	0	0	0	0	4	57	154
	HDD18.3	2080	452	393	304	139	19	1	0	0	3	92	268	408
	CDD10.0	2754	2	3	27	130	324	471	580	587	397	190	39	5
	CDD18.3	1138	0	0	1	9	85	222	322	329	151	19	1	0
	CDH23.3	14024	0	0	6	107	870	2678	4250	4449	1507	155	2	0
	CDH26.7	7313	0	0	0	21	307	1386	2399	2558	622	21	0	0
Wind	WSAvg	4.1	4.1	4.1	4.4	4.1	4.0	4.7	4.6	4.4	4.1	3.6	3.6	3.9
Precipitation	PrecAvg	135	13	13	21	16	12	2	2	2	3	12	20	16
	PrecMax	241	32	49	90	51	141	26	17	29	27	57	86	47
	PrecMin	65	1	0	1	0	0	0	0	0	0	0	0	0
	PrecStd	42	8	9	19	12	21	5	4	5	6	12	19	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	17.2	23.9	29.2	35.1	39.2	40.8	40.9	37.0	28.9	22.8	17.1
		MCWB	8.3	9.2	12.8	14.9	17.3	19.1	19.8	20.0	18.2	14.8	13.9	9.9
	2%	DB	12.1	14.0	20.0	26.2	32.0	37.2	38.8	39.0	34.0	26.9	19.5	14.2
		MCWB	7.4	8.0	10.7	14.4	16.6	19.0	20.3	19.4	17.5	15.1	12.7	9.1
	5%	DB	10.3	12.1	17.2	24.0	30.0	35.2	37.2	37.3	32.1	25.1	17.6	12.2
		MCWB	6.7	7.4	10.1	13.6	16.3	18.8	20.1	19.0	17.5	15.2	12.2	8.5
	10%	DB	8.9	10.2	15.0	21.8	27.9	33.5	35.7	36.0	30.2	23.1	15.8	10.8
		MCWB	6.3	6.5	8.8	12.8	16.1	18.8	20.0	19.1	17.4	14.9	11.6	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	10.2	13.6	16.7	20.3	25.2	26.5	26.4	23.9	20.8	16.3	11.7
		MCDB	12.2	14.7	21.0	25.4	28.3	32.2	32.4	32.5	29.3	24.0	20.0	15.8
	2%	WB	8.3	8.9	11.7	15.6	19.1	22.8	25.2	24.9	22.2	18.8	14.3	10.2
		MCDB	10.9	12.9	17.9	23.6	26.6	30.2	31.7	30.2	27.9	22.8	17.6	12.6
	5%	WB	7.3	7.9	10.5	14.6	18.3	21.6	24.0	23.9	21.0	17.5	13.2	9.2
		MCDB	9.3	10.7	15.9	21.7	25.6	28.8	30.7	29.8	27.0	21.6	16.2	11.5
	10%	WB	6.4	6.9	9.5	13.7	17.6	20.7	23.1	22.9	19.9	16.4	12.0	8.0
		MCDB	8.4	9.6	14.0	19.9	24.8	28.2	29.7	29.7	26.1	20.7	15.0	10.4
Mean Daily Temperature Range	5% DB	MDBR	7.2	8.4	9.2	10.4	11.3	11.6	11.2	11.7	11.6	10.7	8.8	7.5
		MCDBR	8.6	11.3	13.2	13.9	14.0	13.8	13.3	13.7	13.5	12.8	11.1	8.9
		MCWBR	5.6	6.6	6.5	6.0	5.7	5.8	5.5	6.2	6.8	6.9	6.6	5.9
	5% WB	MCDBR	7.4	9.3	11.3	12.3	11.7	11.5	10.7	11.0	11.4	10.2	9.1	7.6
		MCWBR	5.4	6.0	6.0	5.8	5.4	6.0	5.9	6.3	6.7	6.9	6.2	5.7
Clear-Sky Solar Irradiance	taub	0.412	0.454	0.490	0.568	0.547	0.540	0.576	0.542	0.520	0.488	0.440	0.421	
	taud	2.071	1.972	1.921	1.781	1.875	1.912	1.842	1.907	1.921	1.967	2.042	2.049	
	Ebn at Noon	704	731	754	723	753	758	725	736	718	686	665	653	
	Edh at Noon	117	149	174	213	198	191	204	186	172	148	120	112	
All-Sky Solar Radiation	RadAvg	1.84	2.74	3.95	5.18	6.59	7.48	7.06	6.51	5.32	3.54	2.27	1.66	
	RadStd	0.18	0.25	0.30	0.55	0.31	0.18	0.32	0.29	0.22	0.29	0.16	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.24	N/A	N/A	+1.13	N/A	N/A	N/A	N/A	+123	+130
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page